

DONATION MANAGEMENT SYSTEM

A Java Swing Based Desktop Application

1. Introduction

The Donation Management System is a desktop application developed using Java Swing and Object-Oriented Programming (OOP) concepts. The system is designed to manage clothing and item donations efficiently by allowing users to donate items and administrators to manage records.

2. Project Scope

This project focuses on simplifying donation tracking using a file-based system. It provides a structured way to store, view, and manage donated items without requiring a database.

3. Objectives

- To build a user-friendly donation management system
- To apply OOP principles in real-world application
- To implement file handling for data storage
- To develop a role-based system (Admin & Member)

4. System Overview

The system allows two types of users: Admin and Member. Members can donate items while Admin can view and delete donation records.

5. System Features

Member: Donate items, provide personal details, submit donation request

Admin: View all donations, delete records by ID

6. Class Design (OOP Structure)

- User (Abstract Class): Base class for system users
- Admin: Handles administrative operations
- Member: Represents donors
- ClothingItem: Stores donation data
- FileRepository: Handles file operations
- MainFrame: GUI interface using Java Swing

7. File Handling

All donation records are stored in a text file named items.txt.

Format:

id | category | status | donorName | donorEmail | donorPhone

8. OOP Concepts Used

- Encapsulation: Data hiding using private/protected fields
- Inheritance: Admin and Member extend User class
- Abstraction: Abstract User class
- Polymorphism: Method overriding (getRole)

9. Advantages

- Simple and easy-to-use GUI
- Lightweight file-based storage system
- Clear role separation between users

10. Limitations

- No database integration
- Basic UI design
- Limited security features

11. Future Improvements

- Add database (MySQL / SQLite)
- Improve UI design with modern framework
- Add search and filtering system
- Add authentication system with encryption

12. Conclusion

The Donation Management System successfully demonstrates the use of Java Swing and OOP principles. It provides a simple yet effective solution for managing donations and can be expanded into a full-scale application in the future.